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ARTICLE

# Population Genomic Analysis of Brook Trout in Pennsylvania's Appalachian Region

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***Abstract***
**Informed conservation of stream fishes requires detailed understanding of the effects of both natural processes and anthropogenic activities on genetic diversity. Brook Trout *Salvelinus fontinalis*, a salmonid native to eastern North America, typically resides in cold, high-quality stream ecosystems. The species has not only faced historical anthropogenic pressures, but also confronts current and future pressures. In a genetic analysis we used a reduced representation sequencing method (ddRADseq) to characterize 63 individuals from 23 streams where Brook Trout are native in the Appalachian region of Pennsylvania. A total of 2,590 loci passed filtering criteria, and 53% displayed significant association with a major stream drainage basin (Susquehanna or Allegheny; mean $F_{ST}$ = 0.085). Mapping of the sequencing reads to the Atlantic Salmon *Salmo salar* genome revealed no clustering of high interdrainage $F_{ST}$ values to specific genome regions. Evidence for genetic heterogeneity within each drainage basin was also detected. Stepwise regression of observed heterozygosity against geographic and environmental features revealed that drainage basin and effective area of watersheds were significant predictors of observed heterozygosity of Brook Trout within streams. Natural features such as waterfalls and major drainage basin, as well as the effects of dams and acid-mine drainage have fragmented habitat and shaped genetic diversity within Brook Trout populations in the Appalachian region of Pennsylvania, overall indicating the vulnerability of this species to increased industrialization.**

Connectivity of freshwater fishes in the northeastern United States is affected by many natural and anthropogenic environmental challenges such as drainage from abandoned coal mines, hydroelectric dams, and natural gas development (i.e., fracking). Acid-mine drainage (AMD) leads to habitat fragmentation by creating highly toxic, stable, dead zones that limit passage between unaffected regions. Dead zones result from low pH and high levels of heavy metals and sediment in streams (Herlihy et al. 1990; Cole et al. 2001). Over 5,000 km (3,000 mi) of stream waters in Pennsylvania (Pennsylvania Water Science Center, unpublished data) and over 10% of streams in the Appalachian region (Herlihy et al. 1990) are affected by AMD. Hydroelectric dams are another anthropogenic challenge known to fragment aquatic communities by limiting movement of fish upstream or downstream from dams and between watersheds (Morita and Yamamoto 2002). In the past 10–15 years, another potential challenge to fish survival and movement has arisen in the form of fracking. Fracking has grown exponentially in the northeastern United States, and over 60,000 wells are projected to be developed in Pennsylvania alone by 2030 (Johnson 2010). Both withdrawal of water from streams for fracking and effluent spills from fracking have the ability to significantly alter stream physiochemical dynamics (Entrekin et al. 2011; Weltman-Fahs and Taylor 2013) and presumably alter fish passage and connectivity. The effects of such challenges on genetic connectivity of stream fishes within Pennsylvania are largely unknown.

Brook Trout *Salvelinus fontinalis*, a salmonid native to Pennsylvania and other northeastern U.S. states, is a freshwater fish affected by anthropogenic factors. Increasing anthropogenic pressures have eliminated Brook Trout from much of their historic range in the eastern United States, and only 5% of subwatersheds contain intact populations (EBTJV 2013). In Pennsylvania alone, Brook Trout are experiencing

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extirpation and heavy population reduction (>50%) across 73% of their native watersheds due to increased water temperatures, sedimentation, AMD, and fish passage barriers (EBTJV 2013). An understanding of their complex life histories and response to diverse environmental challenges is vital to conservation efforts within their native range.

Previous studies in the northeastern United States and eastern Canada have used genomic data to show correlations between population structure and several environmental variables within Brook Trout populations. Landscape structure has shaped Brook Trout population structure in Maine and Québec where a significant negative relationship has been found between mean heterozygosity and altitude, and significant variance in genetic structure has been explained by drainage basin structure (Castric et al. 2001). Population structure of trout throughout northeastern North America is strongly influenced by glacial history. A single mitochondrial haplotype dominates populations in much of the northern region beyond Maine and into Newfoundland and Labrador, Québec, and Ontario (Danzmann et al. 1998). The same study also found significant heterogeneity in haplotype frequency among populations in different watershed drainage basins in other regions of the species' range. Populations of Brook Trout in Newfoundland and Labrador were highly divergent due to the presence of waterfalls, and genetic divergence increased with waterway distance between streams (Gomez-Uchida et al. 2009). Previous studies have hypothesized that unexplained variance in genomic data among Brook Trout populations may be due to habitat quality influencing population structure and is related to habitat selection by anadromous individuals returning from sea (Rogers and Curry 2004). The role geographical and ecological factors play in trout genetic structure in the mid-Atlantic region of the United States is not well known, particularly in regions with a high prevalence of AMD, hydroelectric dams, or fracking.

Natural populations may be efficiently genetically assayed at large numbers of loci using recently developed genotyping-by-sequencing technologies. Double-digest restriction site associated DNA sequencing (ddRADseq) (Peterson et al. 2012) is one of many recently developed genotyping-by-sequencing (GBS) methods developed to rapidly discover and score thousands of novel single-nucleotide polymorphisms in nonmodel species for collections of individuals. In ddRADseq, genomic DNA is cut with two restriction enzymes, and DNA fragments of a certain size are sequenced with short-read sequencing technologies at one or both fragment ends in multiple individuals. Such GBS technologies have been successfully applied in a diversity of studies involving nonmodel species that includes phylogeography (Emerson et al. 2010), phylogeny reconstruction (Eaton 2014), population structure (Lal et al. 2016), and characterization of sex determination system (Fowler and Buonaccorsi 2016). Compared with traditional batteries of a dozen microsatellite markers, for example, a panel of thousands of single-nucleotide polymorphism (SNP) loci distributed throughout the genome provides an increased resolution of population structure and an ability to find loci under the influence of selection.

In the present study, we used a reduced representation sequencing method (ddRADseq) to genotype native Brook Trout populations within the Appalachian region of Pennsylvania to characterize population genetic diversity and structure in relation to natural and anthropogenic factors. Population connectivity may be reduced by a combination of historical factors such as watershed separation, as well as contemporary impediments to gene flow such as major dams and dead zones caused by AMD. Diversity may also be influenced by population reduction as a result of exposure to fracking effluent.

## METHODS

*Sampling and laboratory analysis.*—We sampled 64 individuals from 23 putative populations of native Brook Trout in central Pennsylvania (Table 1; Figure 1; Supplementary Table S.1 available in the online version of this article). Our general sampling strategy was to target streams for which there was or was not nearby fracking activity and for which there were no direct impacts by AMD (i.e., were not in dead zones); these streams were from two major drainages represented in Pennsylvania, the Allegheny River (leading to the Ohio River) and Susquehanna River drainage (leading to Chesapeake Bay) systems. Only one stream used in this study, Alex Branch, experienced ongoing contamination related to the Marcellus shale natural gas extraction. Between 2009 and 2010, over 2,700 L (7,000 gal) of surfactants, flowback fluid, and well-wash fluid was released into the environment, which negatively impacted the stream ecosystem (Levis 2011). These spills into Alex Branch affected fish communities and reduced Brook Trout numbers to fewer than five individuals captured per year (Grant et al. 2015). We aimed to have representative collections from both the Allegheny and Susquehanna watersheds that would allow us to detect any strong within-basin patterns of differentiation. Pristine "control" streams without downstream impacts do not exist in Pennsylvania.

Samples of DNA were extracted from fin clips using the DNeasy kit and protocol from QIAGEN, and ddRADseq (Peterson et al. 2012) was performed. Samples were randomly allocated to sequencing library pools comprised of 16 individually barcoded fish, and libraries were constructed at Indiana University following Peterson et al. (2012, protocol S1), using an agarose gel size-selection method (range, 200–1,000 bp). Genomic DNA from each individual was cut with two 6-bp recognition site enzymes (*Sph*I and *Eco*RI), as we were aiming for fewer (~5,000–10,000) loci with high coverage with a reference genome of between 500 Mbp and 1 Gbp (Peterson et al. 2012). A total of one lane of 2 × 100-bp, paired-end,

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TABLE 1. Brook Trout sampling locations in the Appalachian region of Pennsylvania with selected metadata including stream, latitude, longitude, number of individuals per stream (*N*), watershed drainage, observed heterozygosity over 2,590 loci, effective watershed area, and obstruction limiting effective watershed area; UNT = unnamed tributary, SBNFRC = South Branch North Fork Redbank Creek.

| Stream | Latitude | Longitude | *N* | Drainage | $H_o$ | $\text{Log}_{10}$ (effective watershed area [km$^2$]) | Obstruction limiting effective watershed area |
|---|---|---|---|---|---|---|---|
| 1. Haven's Run | 41.65 | –78.38 | 3 | Allegheny | 0.271 | 3.75 | Kinzua Resevoir |
| 2. UNT to Clarion | 41.64 | –78.68 | 2 | Allegheny | 0.232 | 3.39 | Piney Dam |
| 3. Long Run | 41.61 | –78.73 | 3 | Allegheny | 0.289 | 3.39 | Piney Dam |
| 4. UNT to Tionesta | 41.61 | –78.96 | 3 | Allegheny | 0.250 | 3.09 | Tionesta Dam |
| 5. Straight Creek | 41.59 | –78.51 | 3 | Allegheny | 0.282 | 2.27 | East Branch Resevoir Dam |
| 6. Camp Run | 41.58 | –79.34 | 2 | Allegheny | 0.264 | 4.38 | Allegheny River[a] |
| 7. Little Wolf Run | 41.54 | –78.69 | 2 | Allegheny | 0.288 | 3.39 | Piney Dam |
| 8. Bear Creek | 41.52 | –78.71 | 2 | Allegheny | 0.300 | 3.39 | Piney Dam |
| 9. North Branch Island Run | 41.40 | –78.75 | 3 | Allegheny | 0.298 | 3.39 | Piney Dam |
| 10. Laurel Run | 41.34 | –78.79 | 3 | Allegheny | 0.255 | 3.39 | Piney Dam |
| 11. Vineyard Run | 41.31 | –78.79 | 3 | Allegheny | 0.255 | 1.00 | AMD |
| 12. SBNFRC | 41.27 | –78.93 | 2 | Allegheny | 0.264 | 2.95 | AMD |
| 13. Ben's Creek | 40.39 | –78.61 | 3 | Allegheny | 0.208 | 1.28 | Small |
| 14. UNT to Driftwood | 41.59 | –78.44 | 3 | Susquehanna | 0.258 | 2.92 | Bennet Branch |
| 15. Alex Branch | 41.18 | –78.42 | 3 | Susquehanna | 0.239 | 1.93 | AMD |
| 16. Deer Creek | 41.16 | –78.30 | 3 | Susquehanna | 0.228 | 1.43 | AMD |
| 17. Dixon Run | 41.16 | –78.39 | 3 | Susquehanna | 0.214 | 1.93 | AMD |
| 18. Cold Stream | 41.13 | –78.42 | 3 | Susquehanna | 0.246 | 1.93 | AMD |
| 19. Stone Run | 41.12 | –78.46 | 3 | Susquehanna | 0.235 | 1.76 | AMD |
| 20. Crooked Run | 41.10 | –78.38 | 3[b] | Susquehanna | 0.230 | 1.93 | AMD |
| 21. Lick Run | 41.10 | –78.47 | 3 | Susquehanna | 0.244 | 1.76 | AMD |
| 22. Dry Gap Run | 40.53 | –78.44 | 3 | Susquehanna | 0.230 | 3.32 | Petersburg Power Dam |
| 23. Diamond Run | 40.33 | –78.58 | 3 | Susquehanna | 0.234 | 3.40 | Raystown Dam |

[a] Subtract Kinzua, Piney, and Tionesta.

[b] One individual discarded due to low read count.

Illumina HiSeq sequencing was performed on the genomic DNA libraries.

*Genotyping.*—The program ref_map.pl version 1.36 in Stacks was used to execute the Stacks pipeline and process the data (Catchen et al. 2013). Raw data were demultiplexed using default settings of the process_radtags script (discards reads with mean phred quality score < 10 using a sliding window 15% the length of the read). One individual from Crooked Run with a very low read count was discarded from analysis, leaving 63 individuals for analysis. Reads were aligned to the Atlantic Salmon *Salmo salar* genome (version AGKD04.1.fsa_nt) using the Bowtie2 algorithm at default settings (Langmead and Salzberg 2012). Between 7% and 13% of read pairs aligned concordantly exactly one time and were retained. During the course of data analysis it became clear that river drainage basin (Allegheny River or Susquehanna Rriver) was the dominant factor driving population genetic structure in the sampled area, so subsequent analysis followed that stratification. Construction of the Stacks catalog, SNP calling, and genotype construction was performed using the ref_map.pl pipeline, with a requirement of 10 reads per locus (i.e., RAD-tag), maximum error probability of 0.10, and default maximum number of mismatches at a locus (parameter *n*) among individuals of 1. The Populations algorithm was used to further filter genotypes, grouped by drainage, for no more than 25% missing data at a locus, a maximum heterozygosity of 0.75 (a theoretical expected maximum of 0.5 is achieved at equal allele frequencies, and this was used to help remove paralogs), minor allele frequency (MAF) of at least 0.05 within group, sampling of a single random SNP per locus, and presence of loci in both groups. While chromosomal rearrangements have occurred between reference and target as well as likely changes in gene function, alignment to reference may reveal clustering of population genetic patterns to regions of the Brook Trout genome that would otherwise be missed. Localization of patterns to certain regions of the genome could provide clues regarding the function of genes related

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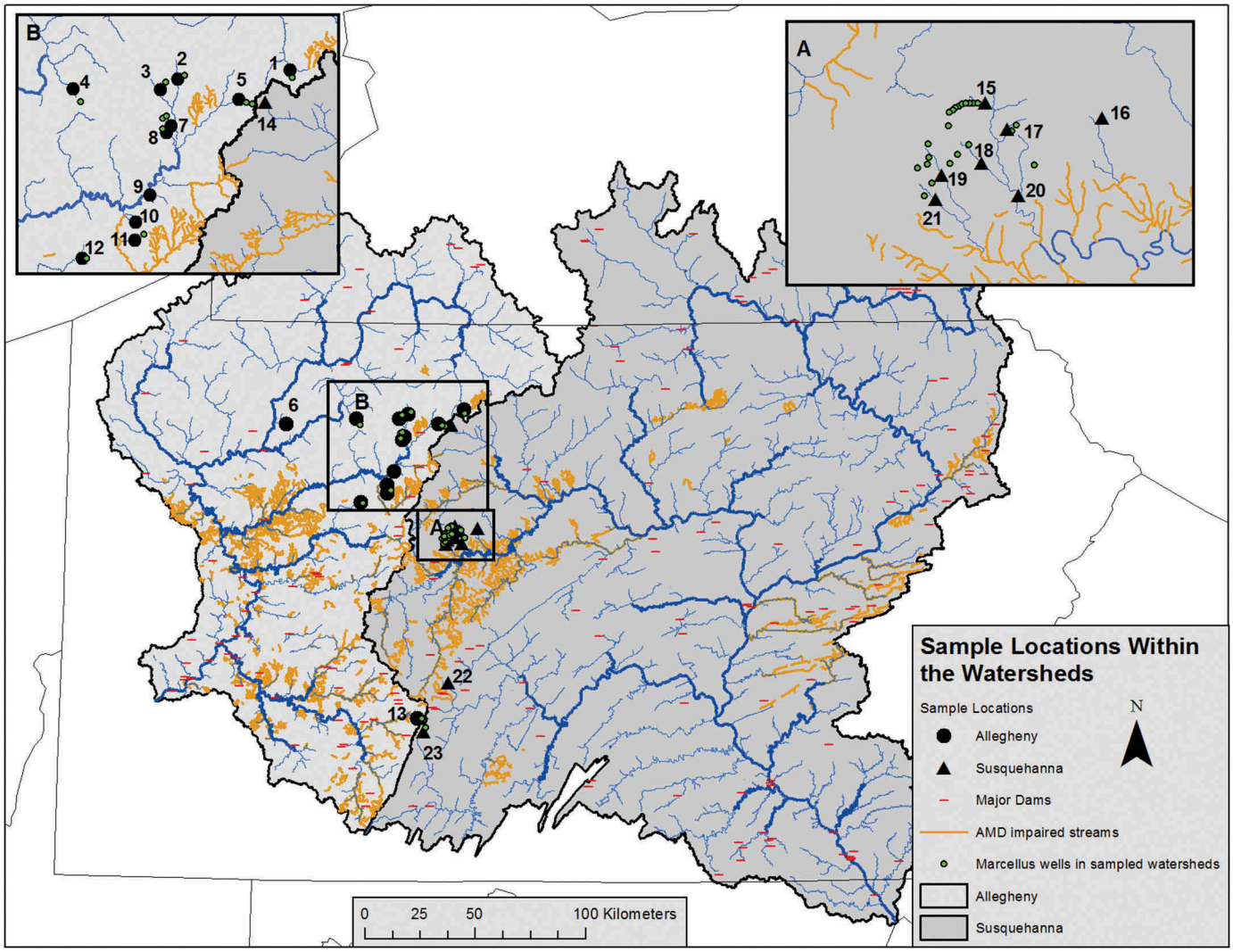


FIGURE 1. Sampling locations for Brook Trout, impaired waterways, and environmental features limiting effective watershed size. Fifth-order streams and higher are shown in heavier blue lines. Stream numbers follow the listing in Table 1.

to the pattern. The Populations component of the Stacks pipeline produced population summary statistics for each drainage ($F_{ST}$, $H_e$ and $H_o$). After locus construction and SNP calling, loci were assessed for deviations from Hardy–Weinberg equilibrium (HWE) within drainage basin using the probability test of Genepop version 4.4.3 (Rousset 2008) and removed if $P < 0.001$. Loci were also assessed for genotypic association with sequencing library pool using the exact test of Genepop version 4.4.3 and removed if $P < 0.001$. This was done because fragment selection methodology can result in differential representation of loci among library pools and SNP allelic identity may be associated with fragment length (Puritz et al. 2014). Loci were tested for selection outliers using Bayescan version 2.1 (Foll and Gaggiotti 2008) and PCAdapt (Duforet-Frebourg et al. 2014). Bayescan identifies selection outliers by comparing models of data with and without a locus-specific selection coefficient estimated and includes control for false discovery rate (FDR; $q$-value < 0.05). Given that a few thousand loci are present in the data, the prior odds of a SNP being neutral was set to 100:1; otherwise default parameters were used. The program PCAdapt performs a Bayesian factor analysis and determines loci that atypically correlate to population structure as determined by the inferred factors. For PCAdapt, Bayes Factors for each SNP summarize strength of evidence that a locus is under selection. Interpretation follows Jeffrey's scale of evidence for Bayes Factors (e.g., >10 indicates strong evidence for the model including selection at a locus).

*Genetic structure.*—STRUCTURE version 2.3.4 was used to evaluate major patterns of population subdivision without using prior population information and used the admixture model, correlated allele frequencies, and five iterations of a burn in of

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10,000 steps followed by 20,000 steps for parameter estimation. Likelihood of the data was assessed for 1–12 possible populations. The maximum population number ($k$) was limited to 12 due to both the low sample size within streams and the clustered nature of our geographic sampling. Also, there is significant annual stream connectivity among several geographically proximate sampling sites, so we anticipated significantly fewer than 23 populations. Population number inference followed the Evanno method (Evanno et al. 2005) and was extracted through STRUCTURE HARVESTER (Earl 2012) and CLUMPP (Jakobsson and Rosenberg 2007) to average over replicates. To assess population structure and relatedness across alleles, we also used principal component analysis (PCA) on individual genotypes as implemented in the ade4 package (Chessel et al. 2004) in the program R (R Development Core Team 2015). Isolation by distance among individuals within the Susquehanna and Allegheny River basins was performed using Mantel tests in Genepop version 4.4.3 with both "a" and "e" genetic distance options (Rousset 2000, 2008) and 10,000 permutations. For Stacks-generated data sets, population genetic summary statistics were obtained using the Populations subroutine. Smoothing of divergence and diversity estimate patterns among and within drainages was performed using a custom script in the program R (see supplementary files 1 and 2 referenced in the Supplement) using a normal kernel following Catchen et al. (2013) with a window size of 150 Kbp and significance assessed over 1,000 replications. To assess significance (FDR $q$-value < 0.05; Benjamini and Hochberg 1995), $F_{ST}$ values were randomly shuffled 1,000 times across the identified SNP locations and smoothed to determine a null distribution of smoothed scores for the known distribution of $F_{ST}$ values. Empirical $p$-values were generated from this distribution and false-discovery corrected using the $q$-value R package. In addition, to identify possible differences in average divergence among scaffolds, we conducted an ANOVA comparing mean $F_{ST}$ value between each scaffold with more than 10 loci. We also checked the correlation between the $F_{ST}$ value at each locus and the $F_{ST}$ value at the next locus on the same chromosome.

*Environmental variables.*—Of the 23 Brook Trout collections, roughly half were located within the Susquehanna River drainage basin ($n$ = 10) and half in the Allegheny River drainage basin ($n$ = 13). Prior to analysis environmental variables with strong right skew were $\log_{10}$ transformed and checked for multicollinearity ($r$ > 0.7). At each sampling point for the 23 collections, predictors included in the association analyses below were drainage basin, number of fracking well pads within watershed, mean stream slope (degrees), percent forest within watershed, Brook Trout abundance, stream pH, stream temperature (°C), $\log_{10}$ (watershed area [ha]), $\log_{10}$ (effective watershed area [km$^2$]), $\log_{10}$ (geographic distance [km] to network midpoint within watershed [see below]), and $\log_{10}$ (AMD impairment distance [km] to watershed network midpoint). A stream was considered to be impaired by AMD if it was listed as impaired for any type of AMD by the Pennsylvania Department of Environmental Protection (PADEP 2015). For each watershed, the number of well pads, mean stream slope, percent forest, and watershed area were assessed using GIS (see the Supplement). Brook Trout abundance was obtained using a standard multipass electrofishing method on a representative 100-m reach for each collection. Stream water temperature and pH were collected during summer low flows with an Oakton multiparameter probe that was calibrated weekly. Watershed delineation, geographic distance along watercourse, effective watershed size, and impairment distance are described in detail in the Supplement. Briefly, effective watershed size was calculated for each sample point in order to examine the effects of movement barriers on available habitat size for Brook Trout populations. Factors that could limit movement for this analysis included impairment from AMD or large dams that restrict fish passage (see the Supplement). Impairment distance for each sample point was calculated to quantify strength of isolation of a stream to the center point of the watershed. This distance was determined by measuring the length of stream that is impaired by AMD between each sample point and a midpoint on the main stem of the respective drainage (see the Supplement). Significance of differences in environmental variables between streams belonging to the two different drainages was assessed with a $t$-test or Kruskal–Wallis nonparametric test depending on variable normality, assessed with Anderson Darling normality test in Minitab (State College, Pennsylvania), and FDR corrected at the 0.05 level. To detect effects of environmental variables on genetic diversity, stepwise regression analyses on observed heterozygosity was performed with α-value of 0.15 to enter or be removed from the model.

## RESULTS

Stacks produced 3,393 loci that passed filters implemented by Populations, including MAF 0.05. Of these loci, HWE deviations were detected at 334 loci (77% of which showed heterozygote excess within at least one basin), and genotypic bias associated with library pool was detected at 469 loci, leaving 2,590 loci for further analysis. Of these, none were identified by Bayescan (all $q$-values > 0.73) or PCAdapt (all SNP Bayes Factors < 2.4) to have strong evidence for selection between drainages. For comparison, only two loci had $q$-values < 0.05 when Bayescan was run in preliminary trials with prior odds of neutrality set to 10:1.

STRUCTURE analysis revealed that while genetic subdivision into three populations had the highest absolute mean likelihood, division into two major populations was supported by the Evanno method (Table 2; Figure 2). Resulting $q$-values were highly concordant among five iterations (mean SD = 0.0004, maximum SD = 0.001). Division was well aligned with drainage basin, although three individuals of predominantly

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TABLE 2. Interpretation of STRUCTURE results following Evanno et al. (2005). Numerical values with the highest Delta *K* are shown in bold italics; NA = not applicable.

| *K* | Repetitions | Mean LnP(*K*) | SE LnP (*K*) | Ln′(*K*) | \|Ln″ (*K*)\| | Delta *K* |
|---|---|---|---|---|---|---|
| | | | **Overall** | | | |
| 1 | 5 | –136,389 | 6 | NA | NA | NA |
| ***2*** | ***5*** | ***–119,020*** | ***7*** | ***17,369*** | ***15,879*** | ***2,134*** |
| 3 | 5 | –117,531 | 42 | 1,489 | 1,540 | 37 |
| 4 | 5 | –117,582 | 73 | –51 | 32 | 0.4 |
| 5 | 5 | –117,665 | 93 | –83 | 15,003 | 161 |
| 6 | 5 | –132,751 | 33,713 | –15,086 | 17,952 | 1 |
| 7 | 5 | –129,884 | 23,913 | 2,866 | 8,999 | 0 |
| 8 | 5 | –118,020 | 859 | 11,865 | 12,444 | 14 |
| 9 | 4 | –118,599 | 1,662 | –580 | 1,728 | 1 |
| 10 | 5 | –120,906 | 3,958 | –2307 | 3,975 | 1 |
| 11 | 5 | –119,239 | 2,477 | 1,667 | 11,522 | 5 |
| 12 | 5 | –129,094 | 15,541 | –9,855 | NA | NA |
| | | | **Susquehanna** | | | |
| 1 | 5 | –50,808 | 10 | NA | NA | NA |
| 2 | 5 | –50,804 | 680 | 3 | 655 | 1 |
| ***3*** | ***5*** | ***–50,146*** | ***88*** | ***658*** | ***947*** | ***11*** |
| 4 | 5 | –50,435 | 216 | –289 | 930 | 4 |
| 5 | 5 | –51,654 | 1,181 | –1,219 | 221 | 0.2 |
| 6 | 5 | –53,094 | 2,343 | –1,440 | NA | NA |
| | | | **Allegheny** | | | |
| 1 | 5 | –69,343 | 11 | NA | NA | NA |
| ***2*** | ***5*** | ***–66,818*** | ***39*** | ***2,525*** | ***3,074*** | ***79*** |
| 3 | 5 | –67,367 | 158 | –549 | 3,970 | 25 |
| 4 | 5 | –71,886 | 6,069 | –4,519 | 3,028 | 0.5 |
| 5 | 5 | –73,377 | 13,297 | –1,491 | 17,096 | 1 |
| 6 | 5 | –91,963 | 48,412 | –18,587 | NA | NA |

Susquehanna River ancestry were present in three different streams within the Allegheny River basin. Credibility intervals (90%) for individual *q*-values for these individuals were on the order of ±4%. Plotting the axes of the PCA showed clear separation of individuals into two drainages, where the same three individuals from the Allegheny River drainage were intermediate or grouped with Susquehanna River drainage (Figure 3). Average $F_{ST}$ between drainages was 0.085, and a total of 1,367 loci (53%) were significantly divergent between drainages after a correction for FDR. However, no evidence for clustering of loci with high interdrainage $F_{ST}$ to specific regions of the Atlantic Salmon genome was detected (Figure S.1 [upper panel]). None of the smoothed scores were significantly higher than expected based on randomization methods ($P > 0.05$). An ANOVA did show an effect of chromosome on mean $F_{ST}$ value ($F = 1.41$, $P = 0.04$); however, a post hoc Tukey honestly significant difference (HSD) test found no significant pairwise difference in $F_{ST}$ values (all *P*-adjusted > 0.15).

Within drainages, exact probability tests of allele frequency heterogeneity among streams revealed evidence of significant divergence, with 1.3% (29 of 2,204) and 1.3% (33 of 2,467) of loci significant after a correction for FDR within the Susquehanna and Allegheny River basins, respectively (Figure S.2). STRUCTURE analysis within drainages also supported population subdivision within each basin (Table 2; Figure 2). Individuals from relatively geographically distant southern streams formed distinct clusters within both Allegheny (Ben's Creek) and Susquehanna (Diamond Run, Dry Gap Run) river drainages. Within the Susquehanna River basin, individuals from Dixon Run formed a third cluster, and a few individuals from the next closest stream (Alex Branch) were the only individuals with *q*-values greater than 0.1 for that cluster. Examination of the PCA (Figure 3) also revealed a grouping of a more southern and geographically distant stream within the Allegheny River basin (Ben's Creek). Correlation of genetic and geographic distance was significant using the Mantel test among individuals within the Susquehanna River drainage using either genetic distance (*a*: $P = 0.0001$; *e*: $P < 0.00001$), but was not significant within the Allegheny River drainage (*a*: $P = 0.067$; *e*: $P = 0.709$), notwithstanding that the median distance of comparison was higher in the Allegheny River (298 km) than in the Susquehanna River branches (156 km). However, examination of geographic versus genetic comparisons within the Susquehanna River drainage revealed a somewhat disjunct distribution of geographic comparisons (Figure S.3), and all genetic distances were negative.

After a correction for FDR, no environmental variables were significantly different between drainage basins. Stepwise regression of $H_o$ revealed that drainage ($F_{1,\ 19} = 6.18$, $P = 0.003$), effective watershed area ($F_{1,\ 19} = 6.33$, $P = 0.021$) and AMD-impaired distance ($F_{1,\ 19} = 2.43$, $P = 0.136$) were significant predictors ($R^2_{adjusted} = 42.8\%$, $R^2_{predicted} = 24.7\%$), although the latter only marginally. In this analysis, the Allegheny River drainage had higher diversity ($H_o = 0.266 \pm 0.0074$, mean ± SE) than did the Susquehanna River drainage ($H_o = 0.236 \pm 0.0038$), and $H_o$ increased with effective drainage size within both drainages (Figure 4). No evidence for genomic clustering of high or low observed heterozygosity was detected (Figure S.1 [lower panel]).

## DISCUSSION

The aim of this study was to characterize population genetic diversity and structure in relation to natural and anthropogenic factors within Brook Trout populations in the Appalachian region of Pennsylvania. All streams sampled contained naturally reproducing, wild Brook Trout populations as designated by their high-quality, coldwater fishery classification by the Pennsylvania Fish and Boat Commission. We found no record of stocking by the commission at these sites; however, some of the receiving water bodies downstream may

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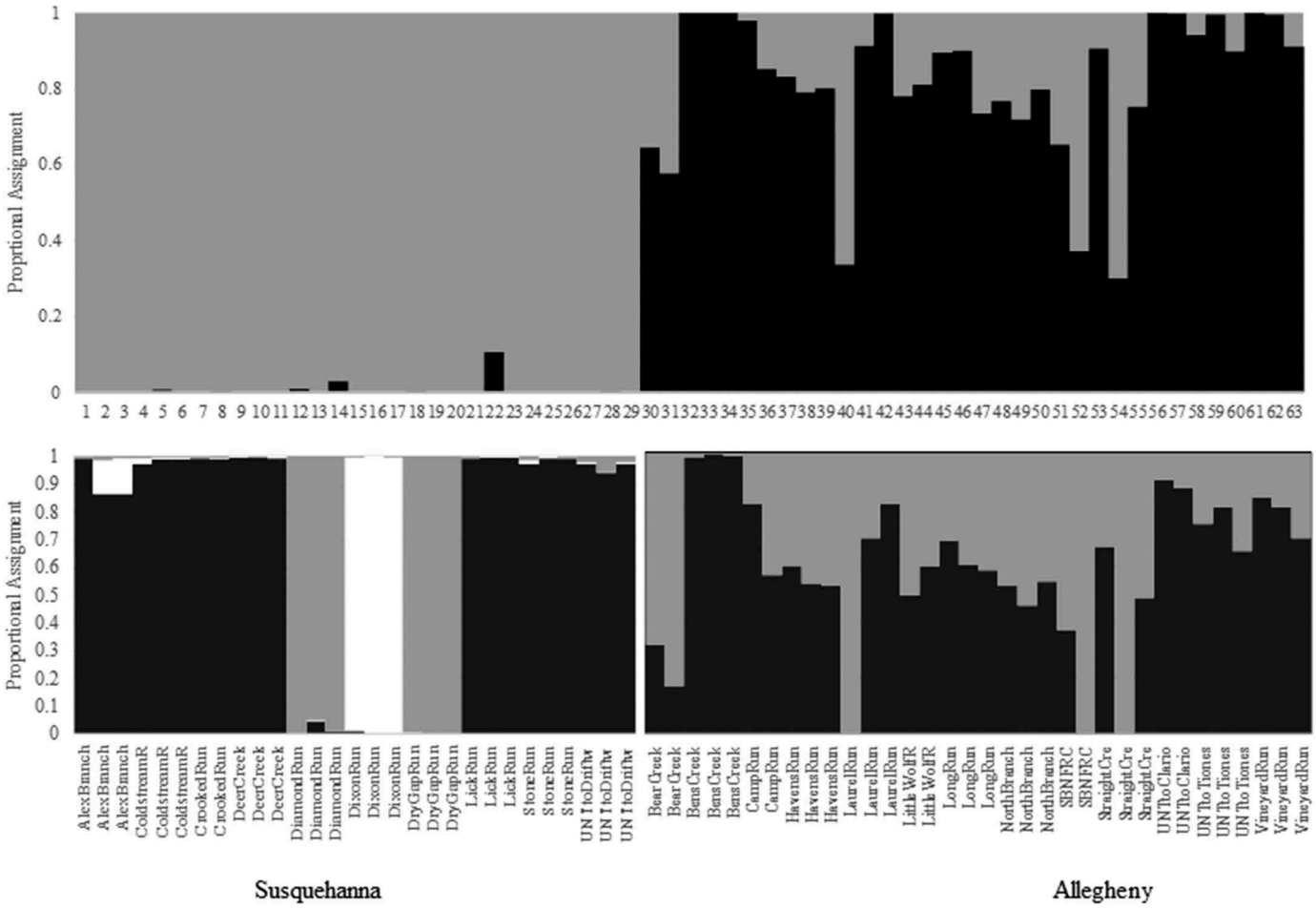


FIGURE 2. STRUCTURE bar plots indicating admixture proportions of Brook Trout individuals to inferred clusters. Top panel shows assignment of all individuals into two clusters from Susquehanna River (1–29) and Allegheny River (30–63) drainages. Bottom panel shows within-drainage STRUCTURE assignment of individuals into three Susquehanna and two Allegheny clusters. Stream origin is indicated.

have been stocked at some previous time, making it difficult to completely eliminate all possibility of genetic influence of hatchery-raised fish. The dominant genetic feature that emerged from our study was the subdivision of samples by the two major watersheds, the Susquehanna and Allegheny rivers. Existence of different lineages within the Ohio River and Chesapeake Bay outlets has been previously established using mitochondrial DNA restriction fragment length polymorphisms (Danzmann et al. 1998). Here we found high fidelity of genetic PCA1 to watershed and genetic divergence that was consistent across the genome and coincided with a gradual process of allopatric population divergence (Via 2009). However, intermediate ancestry of three individuals from the Allegheny River drainage does imply either incomplete lineage sorting, or at least some gene flow between drainages, possibly with a bias in exchange from the Susquehanna River into the Allegheny River drainage. Gene flow between basins may have been facilitated by stocking streams.

Strong evidence for divergence within basins was also detected. It is difficult to interpret whether isolation by distance or distinct groups within the Susquehanna River basin are causing the correlation between genetic and geographic distance as stream distances were clustered. Within both basins relatively distant southern collections formed distinct genetic and geographic clusters. The river distance was several hundred kilometers from these locations to the main group of streams within the basin; this appears to be of sufficient geographic scale for divergence to occur. While in other studies, isolation by riverine distance in this species has been demonstrated in other parts of its range (Gomez-Uchida et al. 2009), it is not clear to what degree this is a naturally continuous process as a result of intrinsically limited dispersal capacity, or one driven by discrete stepped barriers to gene flow such as dams (see Figure 1). Divergence of Dixon Run from nearby streams may be explained by a small population size and isolation from other streams by a relatively high slope that produced a

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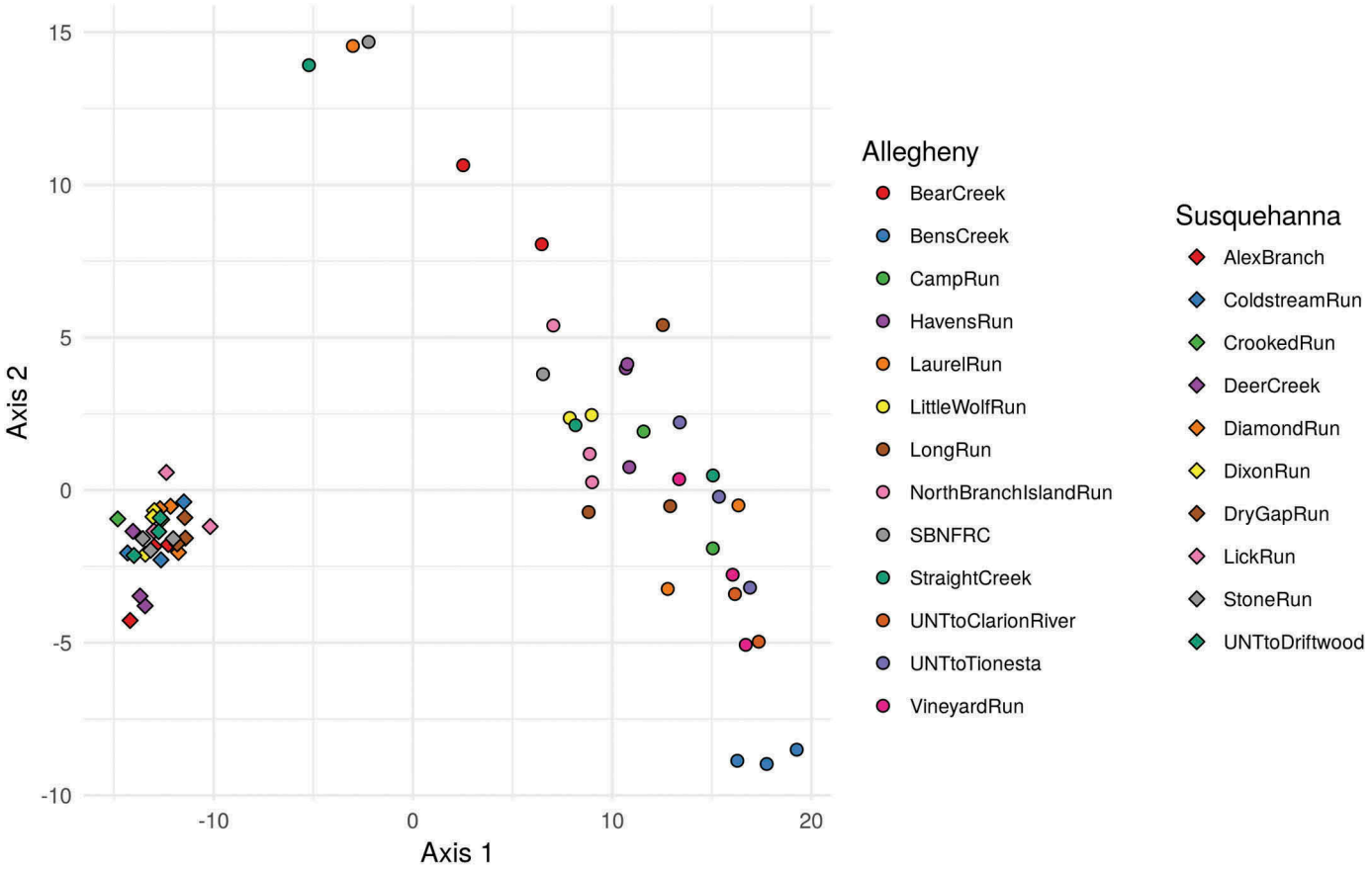


FIGURE 3. PCA axes 1 and 2 including all individual Brook Trout and 2,590 loci from 23 streams in either Allegheny River or Susquehanna River drainage basins in the Appalachian region of Pennsylvania. Scree plots of PCA of the resulting individual genotypes showed that two axes were necessary to describe the data.

series of waterfalls separating this population from downstream populations for much if not all of the year. Waterfalls are known to isolate populations of Brook Trout in other parts of its range (Gomez-Uchida et al. 2009). Relatively high genetic similarity to the Dixon Run cluster within individuals belonging to the next closest sampled stream (Alex Branch) lends further support to the existence of this cluster. Given that there are four fracking well pads in the immediate vicinity for future operations, and that it is a genetically isolated population, it may be considered a relatively vulnerable population. Overall, a stronger signal of divergence within the Susquehanna River basin in combination with lower heterozygosity does suggest a lower effective population size within the drainage despite the overall larger size of the Susquehanna River watershed (Susquehanna River watershed: 71,250 $km^2$; Allegheny River watershed: 30,000 $km^2$). Both AMD and dams have presumably had a larger biological impact on the region, given their presence for over 100 years (approximately 30 Brook Trout generations), in contrast to fracking operations, which are much more recent.

A widespread general divergence between the two watershed drainage basins was detected, characteristic of allopatric population divergence. Little evidence for selective outliers was detected. Sample size and limitations of the drainage basin coverage limit the generality or power of our analysis. Notwithstanding, our results suggest that the divergence detected between these two basins is primarily driven by genetic drift. Selective processes are undoubtedly acting throughout the genome, but may not have been strong or widespread enough either in the genome or geographic region to detect with our experimental design. Alignment to a reference genome from a closely related species may have biased this analysis towards increased coverage of more selectively conserved genomic regions. However, read-mapping of Brook Trout to the Atlantic Salmon genome may be affected by an ancient salmonid genome duplication. At very conserved salmonid regions that have not diverged, Brook Trout reads would map unambiguously more than one time and be discarded. The degree to which bias in read-mapping affects summary statistics is unknown, but the large number of loci supporting major patterns in the present study suggests our

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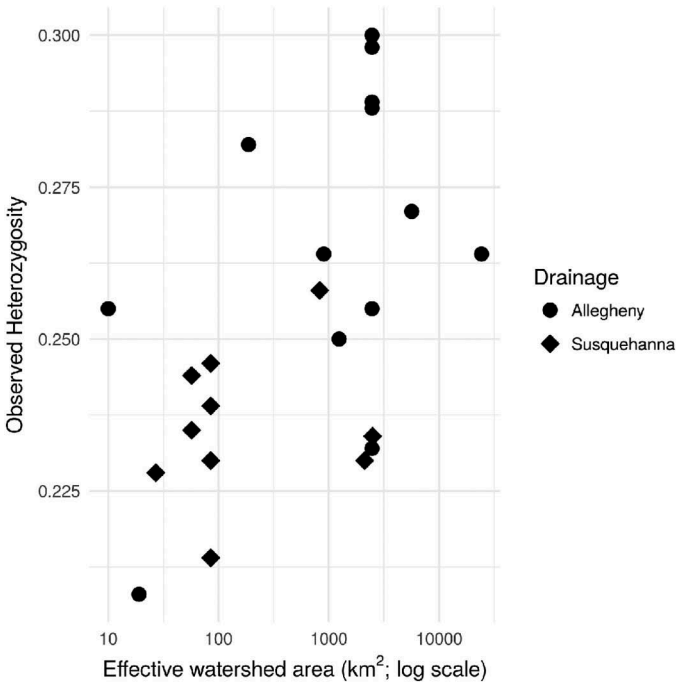


FIGURE 4. Scatterplot of observed Brook Trout heterozygosity versus effective watershed size for 23 stream collections from Allegheny and Susquehanna River drainage basins in the Appalachian region of Pennsylvania. Effective watershed size is described in the text.

major findings are robust. In addition, correlation of trout genetic diversity to size of unaffected regions suggests that increased habitat degradation may cause a greater loss of genetic diversity and reduced ability of the species to adapt to future challenges. The difference in size between adjusted and predicted $R^2$ values in our study, however, suggested some overfitting of our model, indicating that the reported correlates to genetic diversity should be confirmed in future studies. The overall findings of the present study are concordant with other Brook Trout studies showing a low general dispersal capability and an influence of landscape features such as altitude on genetic diversity (Castric et al. 2001) and waterfalls and riverine distance on among-population genetic divergence (Gomez-Uchida et al. 2009). Larger-scale studies within several salmonids have identified genetic–environmental correlations with variables such as drainage pattern, altitude, historical events, precipitation, temperature, distance to the ocean, geology, and subbasin area (Angers et al. 1999; Matala et al. 2011; Olsen et al. 2011; Vincent et al. 2013; Hecht et al. 2015), illustrating that environmental and other features can drive adaptation in that system. Studies at this scale require larger sample sizes to characterize allele frequencies within streams and would be a beneficial next step to further dissect evolutionary interactions within the Brook Trout in the study region.

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## REFERENCES

Angers, B., P. Magnan, M. Plante, and L. Bernatchez. 1999. Canonical correspondence analysis for estimating spatial and environmental effects on microsatellite gene diversity in brook charr (*Salvelinus fontinalis*). Molecular Ecology 8:1043–1053.

Benjamini, Y., and Y. Hochberg. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the Royal Statistical Society Series B (Methodological) 57:289–300.

Castric, V., F. Bonney, and L. Bernatchez. 2001. Landscape structure and hierarchical genetic diversity in brook charr, *Salvelinus fontinalis*. Evolution 55:1016–1028.

Catchen, P., S. Hohenlohe, A. Bassham, et al. 2013. Stacks: an analysis tool set for population genomics. Molecular Ecology 22:3124–3140.

Chessel, D., A. Dufour, and J. Thioulouse. 2004. The ade4 package. R News 4:5–10.

Cole, M., D. Arnole, B. Watten, and W. Krise. 2001. Haematological and physiological responses of brook charr, to untreated and limestone-neutralized acid mine drainage. Journal of Fish Biology 59:79–91.

Danzmann, R. G., R. P. Morgan II, M. W. Jones, L. Bernatchez, and P. E. Ihssen. 1998. A major sextet or mitochondrial DNA phylogenetic assemblages extant in eastern North American Brook Trout *Salvelinus fontinalis*: distribution and postglacial dispersal patterns. Canadian Journal of Zoology 76:1300–1318.

Duforet-Frebourg, N., E. Bazin, and M. G. Blum. 2014. Genome scans for detecting footprints of local adaptation using a Bayesian factor model. Molecular Biology and Evolution 31:2483–2495.

Earl, D. 2012. STRUCTURE HARVESTER: a website and program for visualizing STRUCTURE output and implementing the Evanno method. Conservation Genetics Resources 4:359–361.

Eaton, D. A. 2014. PyRAD: assembly of de novo RADseq loci for phylogenetic analyses. Bioinformatics 30:1844–1849.

EBTJV (Eastern Brook Trout Joint Venture). 2013. Eastern Brook Trout: status and threats. Conservation strategy report. Available: http://easternbrooktrout.org/reports/eastern-brook-trout-status-and-threats/at_download/file. (February 2017).

Emerson, K. J., C. R. Merz, J. M. Catchen, P. A. Hohenlohe, W. A. Cresko, W. E. Bradshaw, and C. M. Holzapfel. 2010. Resolving postglacial phylogeography using high-throughput sequencing. Proceedings of the National Academy of Sciences of the USA 107:16196–16200.

Entrekin, S., M. Evans-White, B. Johnson, and E. Hagenbuch. 2011. Rapid expansion of natural gas development poses a threat to surface waters. Frontiers in Ecology and the Environment 9:503–511.

15488659, 2017, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1080/00028487.2017.1285351 by Ohio State University University Libraries, Wiley Online Library on [26/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Evanno, G., S. Regnaut, and J. Goudet. 2005. Detecting the number of clusters of individuals using the software STRUCTURE: a simulation study. Molecular Ecology 14:2611–2620.

Foll, M., and O. E. Gaggiotti. 2008. A genome scan method to identify selected loci appropriate for both dominant and codominant markers: a Bayesian perspective. Genetics 180:977–993.

Fowler, B. L., and V. P. Buonaccorsi. 2016. Genomic characterization of sex-identification markers in *Sebastes carnatus* and *S. chrysomelas* rockfishes. Molecular Ecology 25:2165–2175.

Gomez-Uchida, D., T. Knight, and D. Ruzzante. 2009. Interaction of landscape and life history attributes on genetic diversity, neutral divergence and gene flow in a pristine community of salmonids. Molecular Ecology 18:4854–4869.

Grant, C. J., A. B. Weimer, N. K. Marks, E. S. Perow, J. M. Oster, and K. M. Brubaker, R. V. Trexler, C. M. Solomon, and R. Lamendella. 2015. Marcellus and mercury: assessing potential impacts of unconventional natural gas extraction on aquatic ecosystems in northwestern Pennsylvania. Journal of Environmental Science and Health Part A: Toxic/Hazardous Substances and Environmental Engineering 50:482–500.

Hecht, B. C., A. P. Matala, J. E. Hess, and S. R. Narum. 2015. Environmental adaptation in Chinook Salmon *Oncorhynchus tshawytscha* throughout their North American range. Molecular Ecology s24:5573–5595.

Herlihy, A., P. Kaufmann, M. Mitch, and D. Brown. 1990. Regional estimates of acid mine drainage impact on streams in the mid-Atlantic and south-eastern United States. Water, Air, and Soil Pollution 50:91–107.

Jakobsson, M., and N. A. Rosenberg. 2007. CLUMPP: a cluster matching and permutation program for dealing with label switching and multimodality in analysis of population structure. Bioinformatics 23:1801–1806.

Johnson, N. 2010. Pennsylvania energy impacts assessment. Nature Conservancy, Harrisburg, Pennsylvania. Available: http://www.marcellus.psu.edu/resources/PDFs/tncenergy.pdf. (January 2014).

Lal, M. M., P. C. Southgate, D. R. Jerry, and K. R. Zenger. 2016. Fishing for divergence in a sea of connectivity: the utility of ddRADseq genotyping in a marine invertebrate, the black-lip pearl oyster *Pinctada margaritifera*. Marine Genomics 25:57–68.

Langmead, B., and S. Salzberg. 2012. Fast gapped-read alignment with Bowtie 2. Nature Methods 9:357–359.

Levis, E. 2011. Texas company pays $93,710 settlement for polluting Clearfield County Creek. Pennsylvania Fish and Boat Commission, Harrisburg, Pennslyvania. Available: https://hero.epa.gov/hero/index.cfm/reference/details/reference_id/2447894. (May 2016).

Matala, A. P., J. E. Hess, and S. R. Narum. 2011. Resolving adaptive and demographic divergence among Chinook Salmon populations in the Columbia River basin. Transactions of the American Fisheries Society 140:783–807.

Morita, K., and S. Yamamoto. 2002. Effects of habitat fragmentation by damming on the persistence of stream-dwelling charr populations. Conservation Biology 16:1318–1323.

Olsen, J. B., P. A. Crane, B. G. Flannery, K. Dunmall, W. D. Templin, and J. K. Wenburg. 2011. Comparative landscape genetic analysis of three Pacific salmon species from subarctic North America. Conservation Genetics 12:223–241.

PADEP (Pennsylvania Department of Environmental Protection). 2015. Integrated list non-attaining. Available: www.pasda.psu.edu. (February 2016).

Peterson, B. K., J. N. Weber, E. H. Kay, H. S. Fisher, and H. E. Hoekstra. 2012. Double digest RADseq: an inexpensive method for de novo SNP discovery and genotyping in model and non-model species. PLOS (Public Library of Science) ONE [online serial] 7:e37135.

Puritz, J. B., M. V. Matz, R. J. Toonen, J. N. Weber, D. I. Bolnick, and C. E. Bird. 2014. Demystifying the RAD fad. Molecular Ecology 23:5937–5942.

R Development Core Team. 2015. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna.

Rogers, S., and R. Curry. 2004. Genetic population structure of Brook Trout inhabiting a large river watershed. Transactions of the American Fisheries Society 133:1138–1149.

Rousset, F. 2000. Genetic differentiation between individuals. Journal of Evolutionary Biology 13:58–62.

Rousset, F. 2008. Genepop'007: a complete reimplementation of the Genepop software for Windows and Linux. Molecular Ecology Resources 8:103–106.

Via, S. 2009. Natural selection in action during speciation. Proceedings of the National Academy of Sciences of the USA 106:9939–9946.

Vincent, B., M. Dionne, M. P. Kent, S. Lien, and L. Bernatchez. 2013. Landscape genomics in Atlantic Salmon *Salmo salar*: searching for gene–environment interactions driving local adaptation. Evolution 67:3469–3487.

Weltman-Fahs, M., and J. M. Taylor. 2013. Hydraulic fracturing and Brook Trout habitat in the Marcellus Shale region: potential impacts and research needs. Fisheries 38:4–15.

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